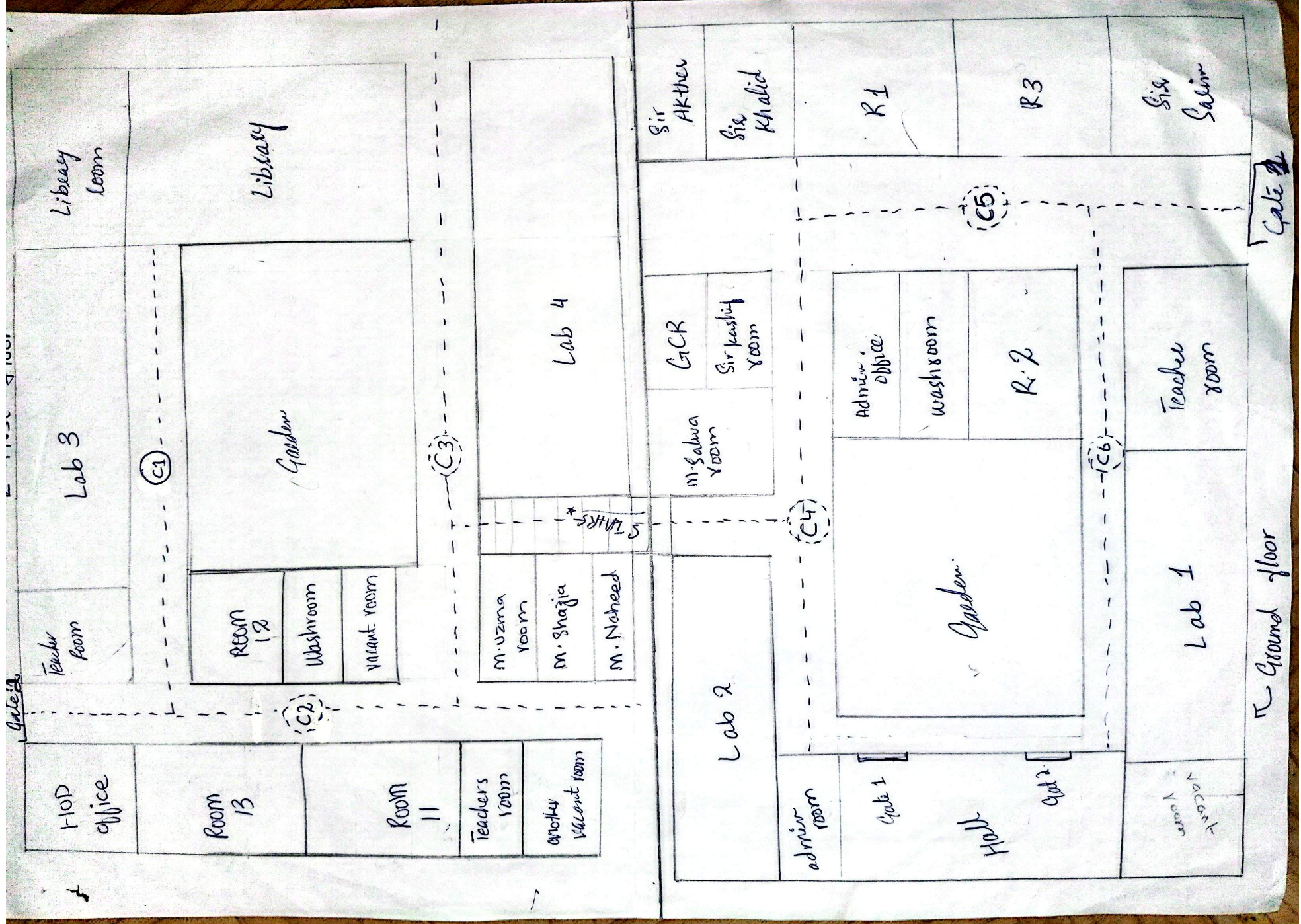




BFS PROBLEM

⇒ LIB TO HALL:-

⇒ Shortest path :-

libroom → c1 → c3 → c2 → c3 → stairs →
c4 → Hall.

initial state → library.

goal state → Hall

path cost → 1

→ c1, c2, c3, c4, c5, c6, are corridors of both
floor combined

→ libroom is the room attached to library
(initial state)

⇒ Enqueue & dequeue :-

Q → [libroom]

V → [libroom]

dequeue libroom and expand.

Q → [c1]

V → [libroom, c1]

dequeue C1 and expand

Q $\rightarrow$ [C2]

V $\rightarrow$ [libroom, C1, C2]

dequeue C2 and expand

Q $\rightarrow$ [C3]

V $\rightarrow$ [libroom, C1, C2, C3]

dequeue C3 and expand

Q $\rightarrow$ [stairs]

V $\rightarrow$ [libroom, C1, C2, C3, stairs]

dequeue stairs and expand

Q $\rightarrow$ [C4]

V $\rightarrow$ [libroom, C1, C2, C3, stairs, C4]

dequeue C4 and expand

Q $\rightarrow$ [C5, Hall]

V $\rightarrow$ [libroom, C1, C2, C3, stairs, C4, Hall]